

Your Name:

ID #:

Worksheet: Limits to Infinity-Part 2

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Show all the work to find the following limits.

(a) $\lim_{x \rightarrow -\infty} (e^{3x} - e^{2x})$

Solution:

$$\begin{aligned}\lim_{x \rightarrow -\infty} (e^{3x} - e^{2x}) &= \lim_{x \rightarrow -\infty} e^{2x}(e^x - 1) \\ &= \lim_{x \rightarrow -\infty} e^{2x} \cdot \lim_{x \rightarrow -\infty} (e^x - 1) = 0 \cdot (-1) = 0\end{aligned}$$

(b) $\lim_{x \rightarrow \infty} \sin\left(\frac{x^2}{2x^2 + x}\right)$

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^2}{2x^2 + x} &= \lim_{x \rightarrow \infty} \frac{x^2}{x(2x + 1)} \\ &= \lim_{x \rightarrow \infty} \frac{x}{2x + 1} = \frac{1}{2}, \text{ then} \\ \lim_{x \rightarrow \infty} \sin\left(\frac{x^2}{2x^2 + x}\right) &= \sin\left(\lim_{x \rightarrow \infty} \frac{x^2}{2x^2 + x}\right) = \frac{1}{2}\end{aligned}$$

(c) $\lim_{x \rightarrow \infty} (\sin x)e^{-x}$

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} (\sin x)e^{-x} &= \lim_{x \rightarrow \infty} \frac{\sin x}{e^x}. \\ -1 &\leq \sin x \leq 1 \implies -\frac{1}{e^x} \leq \frac{\sin x}{e^x} \leq \frac{1}{e^x}, \\ \lim_{x \rightarrow \infty} -\frac{1}{e^x} &= \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0 \\ \implies \lim_{x \rightarrow \infty} \frac{\sin x}{e^x} &= 0 \text{ by the Squeeze Theorem,} \\ \text{so } \lim_{x \rightarrow \infty} (\sin x)e^{-x} &= 0\end{aligned}$$

(d) $\lim_{x \rightarrow \infty} x \sin x$

Solution:

As $x \rightarrow \infty$, x grows without bounds and $\sin x$ oscillates between 1 and -1; so as $x \rightarrow \infty$, $x \sin x$ oscillates large in both positive and negative directions. Therefore, the Limit does not exist.

(e) $\lim_{x \rightarrow \infty} x \ln \frac{1}{x}$

Solution:

$$\ln \frac{1}{x} = -\ln x; \text{ So, } \lim_{x \rightarrow \infty} x \ln \frac{1}{x} = -\lim_{x \rightarrow \infty} x \ln x = -\infty$$

(f) $\lim_{x \rightarrow -\infty} x e^{\frac{1}{x}}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow -\infty} x e^{\frac{1}{x}} \\ &= \lim_{x \rightarrow -\infty} x \cdot \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} \\ &= -\infty \cdot e^{\lim_{x \rightarrow -\infty} \frac{1}{x}} = -\infty \cdot e^0 = -\infty \end{aligned}$$

2. Show all the work to find the following Limits:

(a) $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

Solution:

Let $Q = x^{\frac{1}{x}}$, then we need to find $\lim_{x \rightarrow \infty} Q$.

$$Q = x^{\frac{1}{x}} \implies \ln Q = \frac{1}{x} \ln x$$

In order to find $\lim_{x \rightarrow \infty} Q$, let's find $\lim_{x \rightarrow \infty} \ln Q = \lim_{x \rightarrow \infty} \frac{1}{x} \ln x$ first:

By L'Hôpital's rule,

$$\lim_{x \rightarrow \infty} \frac{1}{x} \ln x = \lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{\text{L.H}}{=} \lim_{x \rightarrow \infty} \frac{(\ln x)'}{(x)'} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

that is to say, $\lim_{x \rightarrow \infty} \ln Q = 0$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \ln Q &= 0 \\ \implies \ln(\lim_{x \rightarrow \infty} Q) &= 0 \\ \implies \lim_{x \rightarrow \infty} Q &= e^0 = 1 \\ \implies \lim_{x \rightarrow \infty} x^{\frac{1}{x}} &= 1 \end{aligned}$$

(b) $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{1}{x}}$

Solution:

Let $Q = \left(\frac{1}{x}\right)^{\frac{1}{x}}$.

$$\begin{aligned} Q &= \left(\frac{1}{x}\right)^{\frac{1}{x}} \\ \implies \ln Q &= \frac{1}{x} \ln \frac{1}{x} = \frac{1}{x} \ln(x^{-1}) = -\frac{1}{x} \ln x \\ \implies \lim_{x \rightarrow \infty} \ln Q &= -\lim_{x \rightarrow \infty} \frac{1}{x} \ln x = -\lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{\text{L.H}}{=} -\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0 \\ \implies \ln(\lim_{x \rightarrow \infty} Q) &= 0 \\ \implies \lim_{x \rightarrow \infty} Q &= e^0 = 1 \\ \implies \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{1}{x}} &= 1 \end{aligned}$$

(c) $\lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^x$

Solution:

Let $Q = \left(\frac{x}{x+1}\right)^x$.

$$\begin{aligned} Q &= \left(\frac{x}{x+1}\right)^{\frac{1}{\frac{1}{x}}} \\ \Rightarrow \ln Q &= x \ln \frac{x}{x+1} = \frac{\ln \frac{x}{x+1}}{\frac{1}{x}} \\ \Rightarrow \lim_{x \rightarrow \infty} \ln Q &= \lim_{x \rightarrow \infty} \frac{\ln \frac{x}{x+1}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{\ln(1 - \frac{1}{x+1})}{\frac{1}{x}} \stackrel{\text{L.H}}{=} \lim_{x \rightarrow \infty} \frac{[\ln(1 - \frac{1}{x+1})]'}{(\frac{1}{x})'} \end{aligned}$$

Now we find $[\ln(1 - \frac{1}{x+1})]'$ and $(\frac{1}{x})'$.

- $[\ln(1 - \frac{1}{x+1})]' = \frac{1}{1 - \frac{1}{x+1}} \cdot (1 - \frac{1}{x+1})' = \frac{1}{1 - \frac{1}{x+1}} \cdot \frac{1}{(x+1)^2} = \frac{1}{(x+1)^2 - (x+1)}$
- $(\frac{1}{x})' = -\frac{1}{x^2}$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{[\ln(1 - \frac{1}{x+1})]'}{(\frac{1}{x})'} = \lim_{x \rightarrow \infty} \frac{\frac{1}{(x+1)^2 - (x+1)}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} -\frac{x^2}{(x+1)^2 - (x+1)} = -1$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln Q = -1$$

$$\Rightarrow \ln(\lim_{x \rightarrow \infty} Q) = -1$$

$$\Rightarrow \lim_{x \rightarrow \infty} Q = e^{-1} = \frac{1}{e}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^x = \frac{1}{e}$$